

# Internship in Physical Organic Chemistry

## Week 3, Part 1: Setup and installation

Prepared by Dr. Alexandra Tsybizova

### Goals for this part:

- Understand what you will compute this week and why
- Install the tools required for the calculations
- Learn the basic terminal commands you will use throughout the week
- Set up your folder structure and verify the complete installation

## 1 What you will do this week and why

In Week 2 you recorded MS/MS spectra at different collision energies and identified the retro-Diels–Alder (RDA) fragments of your compound. You observed that at low energies the precursor ion survives intact, and at high energies it breaks apart. In Week 4 you will analyze the energy barrier for this fragmentation experimentally by constructing breakdown curves. This week you will calculate the same barrier using a computer.

The calculation works as follows. A computer program can calculate the **energy of a molecule** given the positions of its atoms — it tells you how much energy is stored in that particular arrangement. By calculating this energy at many different geometries — from the intact precursor ion, through structures where the two C–ring bonds are progressively stretched, all the way to the two separated fragments — you obtain a curve called a **potential energy profile**. This curve rises from the reactant, passes through a maximum, then descends toward the products. The height of that maximum above the starting point is the **computed energy barrier**.

At the end of the week you will have one number from the computation. In Week 4 you obtain a second number from the experiment. Comparing these two values is the scientific point of the project.

### How does the program calculate the energy?

A molecule's energy depends on how its electrons are distributed around the nuclei. Calculating this exactly is too slow for molecules the size of your flavonoid. The program you will use — called **xTB** — uses a fast approximate method that gives chemically meaningful results and runs in minutes on a laptop. The approximation means the numbers will not be perfectly accurate, but they will be close enough for a meaningful comparison with your experimental result.

## 2 Tools you will install

Five tools are needed this week:

- **Miniconda** — installs and manages scientific software on your laptop. It also provides the terminal window from which you will run all other tools.
- **OpenBabel** — converts a SMILES string (a text code for a chemical structure, explained in Section 3) into a file containing the three-dimensional coordinates of the molecule.

- **xTB** — performs the quantum chemical calculations: optimises molecular geometries and computes the potential energy at each point along the reaction coordinate.
- **Chemcraft** (Windows) or **Avogadro** (macOS) — a molecular visualiser. You will use it to inspect 3D structures and identify atom numbers needed to set up the scan.
- **IBOView** — an orbital visualiser. You will use it to examine how the electron distribution changes along the reaction path.

Allow approximately one hour for the complete installation. Follow each step exactly and verify it before moving on.

### 3 SMILES strings

A **SMILES string** is a compact text representation of a chemical structure. Each atom is written as its element symbol. Single bonds are implicit (not written), double bonds are written as =, and rings are indicated by numbers marking where the ring opens and closes. Hydrogen atoms are not written — they are added automatically.

#### Examples.

Ethanol ( $\text{CH}_3\text{CH}_2\text{OH}$ ) is written as CCO: two carbons followed by one oxygen, all connected by single bonds.

Benzene ( $\text{C}_6\text{H}_6$ ) is written as c1ccccc1: six aromatic carbons in a ring, where the 1 marks where the ring starts and closes.

You will not write SMILES strings by hand. In Part 3 you will look up your compound on PubChem and copy the SMILES from there.

### 4 The XYZ coordinate file

All 3D structures in this workflow are stored in **XYZ files** (extension `.xyz`). The format has three parts: the first line gives the total number of atoms; the second line is a comment that can be blank but must exist; and each remaining line gives the element symbol followed by the  $x$ ,  $y$ ,  $z$  coordinates of one atom in Ångströms.

**Example.** Water ( $\text{H}_2\text{O}$ ), three atoms:

```
3
O  0.000  0.000  0.000
H  0.757  0.586  0.000
H -0.757  0.586  0.000
```

The second line is blank but must be present. A missing second line is the most common cause of errors when running xTB.

Atoms are numbered from 1 in the order they appear in the file. In the example above, O is atom 1, the first H is atom 2, the second H is atom 3. You will need these numbers in Part 4 to tell xTB which bonds to stretch.

## 5 The terminal and basic commands

All calculations this week are run by typing commands into a terminal window.

**Windows:** after installing Miniconda, you will use the **Anaconda Prompt**, which appears in the Start menu.

**macOS:** you will use the **Terminal**, found in Applications → Utilities.

When you open either, you see a line ending with `>` (Windows) or `$` (macOS). This is the prompt, waiting for a command. Type the command and press **Enter** to run it.

Commands you will use:

| Action               | Windows                          | macOS                         |
|----------------------|----------------------------------|-------------------------------|
| Create a folder      | <code>mkdir name</code>          | <code>mkdir name</code>       |
| Move into a folder   | <code>cd name</code>             | <code>cd name</code>          |
| Move one level up    | <code>cd ..</code>               | <code>cd ..</code>            |
| List folder contents | <code>dir</code>                 | <code>ls</code>               |
| Print file contents  | <code>type file.out</code>       | <code>cat file.out</code>     |
| Search inside a file | <code>findstr "text" file</code> | <code>grep "text" file</code> |
| Copy a file          | <code>copy source dest</code>    | <code>cp source dest</code>   |

You will also use `>` to save a calculation's output to a file:

```
xtb molecule.xyz --opt > result.out
```

Without `> result.out` the output scrolls past and is lost. Always redirect output to a file.

**Important:** Never use spaces in file or folder names. Use underscores instead — for example `week3_neutral`, not `week3 neutral`. Spaces cause errors that are difficult to diagnose.

## 6 Folder structure for this week

xTB always writes output files with the same names (`xtbopt.xyz`, `xtbrestart`, and others). If you run two calculations in the same folder, the second overwrites the first. Use a separate subfolder for each calculation.

Create the following structure now, using File Explorer (Windows) or Finder (macOS), or the `mkdir` command:

```
week3/
  practice/      <- Part 2 exercises
  neutral/       <- optimisation of the neutral compound
  ion/           <- optimisation of candidate ions
  scan/          <- potential energy scan
```

Never delete or rename files inside a finished calculation folder.

## 7 Installing Miniconda

1. Go to <https://conda.io/en/latest/miniconda.html>

2. Download the installer for your system:

**Windows:** download **Miniconda3 Windows 64-bit** (the `.exe` file).

**macOS:** download the `.pkg` installer. If your Mac has an M-series chip (M1, M2, M3...), choose **arm64**. If it has an Intel chip, choose **x86\_64**. If you are unsure, go to Apple menu → About This Mac.

3. Run the installer and follow the default options throughout.

**Windows only:** when asked, do **not** tick “Add Miniconda3 to my PATH” and do **not** tick “Register Miniconda3 as default Python”.

4. Open your terminal:

**Windows:** find **Anaconda Prompt** in the Start menu.

**macOS:** open a new Terminal window (Applications → Utilities → Terminal).

You should see `(base)` at the left of the prompt.

Verify:

```
conda --version
```

Expected: `conda 23.x.x` or higher. If you see an error, restart the terminal and try again.

## 8 Creating the environment and installing OpenBabel

A conda **environment** is an isolated workspace for one project, separate from everything else on your computer. You will create one called `physorg`.

Type these commands one at a time, pressing Enter after each. When asked **Proceed** (`[y]/n`)?, type `y`:

```
conda create -n physorg python=3.10
conda activate physorg
conda config --add channels conda-forge
conda config --set channel_priority strict
conda install openbabel
```

The download takes a few minutes.

Type `conda activate physorg` at the start of every new terminal session before running any other command. The prompt shows `(physorg)` on the left when the environment is active.

Verify:

```
obabel --version
```

Expected: `Open Babel 3.x.x`.

## 9 Installing xTB

**macOS:** install via conda and skip to Section 10:

```
conda install xtb
```

**Windows:** the conda version of xTB has a numerical bug on Windows and must not be used. Install the official binary from GitHub instead.

1. Go to <https://github.com/grimme-lab/xtb/releases/latest>
2. Download the file named `xtb-6.x.x-windows-x86_64.zip`
3. Extract the archive to `C:\xtb\`. After extraction, confirm that the following file exists (the version number in the path matches what you downloaded):

```
C:\xtb\xtb-6.7.1\bin\xtb.exe
```

4. In the Anaconda Prompt, add xTB to the search path (replace 6.7.1 with your version):

```
set PATH=C:\xtb\xtb-6.7.1\bin;%PATH%
```

**Windows:** the `set PATH` command is temporary. It must be typed at the start of every new Anaconda Prompt session, immediately after `conda activate physorg`.

Verify (both systems):

```
xtb --version
```

Expected: a banner showing `xtb version 6.x.x`.

## 10 Installing a molecular visualiser

You will use a molecular visualiser to open XYZ files, inspect 3D structures, and identify atom numbers by clicking on atoms.

**Windows — Chemcraft:**

1. Go to <https://www.chemcraftprog.com> and click **Download**
2. Run the installer and follow the default options
3. Open Chemcraft to verify it launches correctly

Chemcraft provides 150 free launches, sufficient for this internship.

**macOS — Avogadro:**

1. Go to <https://avogadro.cc> and download **Avogadro 2** for macOS
2. Open the `.dmg` file and drag Avogadro to your Applications folder
3. Open Avogadro to verify it launches correctly

**Reading atom numbers in your visualiser.**

**Chemcraft (Windows):** open an XYZ file via File → Open. Click on any atom. The status bar at the bottom shows the element symbol and atom number (for example, C14).

**Avogadro (macOS):** open an XYZ file via File → Open. Go to View → Set Label → Atom Index to display atom numbers on all atoms in the 3D view.

**Important:** Avogadro uses **0-based** numbering (atoms are numbered 0, 1, 2...), while xTB uses **1-based** numbering (atoms are numbered 1, 2, 3...). When you identify an atom number in Avogadro, add 1 before writing it into your `scan.inp` file.

## 11 Installing IBOView

IBOView is a standalone orbital visualiser requiring no installation.

1. Go to <http://www.iboview.org/index.html>
2. Download the version for your operating system
3. Extract the archive and place the executable somewhere convenient
4. Double-click the executable to verify it opens

## 12 Verifying the complete installation

Open your terminal and run the startup commands.

### Windows:

```
conda activate physorg
set PATH=C:\xtb\xtb-6.7.1\bin;%PATH%
```

### macOS:

```
conda activate physorg
```

Create a test folder and generate a 3D structure of ethanol:

```
mkdir test_install
cd test_install
obabel -:"CCO" --gen3d -O test.xyz
```

Open `test.xyz` in a text editor and confirm it contains 9 lines of coordinates after a two-line header. Open it in Chemcraft or Avogadro and confirm a recognisable ethanol structure appears.

Run a geometry optimisation (takes a few minutes):

```
xtb test.xyz --opt --chrg 0 --uhf 0 --gfn 2 --norestart > opt.out
```

Check for normal termination:

### Windows:

```
type opt.out | findstr "normal termination"
```

**macOS:**

```
grep "normal termination" opt.out
```

Expected: `normal termination` of `xtb`. Open `xtbopt.xyz` in your visualiser and confirm the optimised structure looks reasonable.

**Windows only:** if the geometry in the visualiser looks exploded (atoms far apart, no bonds), the wrong xTB binary is being used. Run `where xtb` to check which binary is active and contact your supervisor.

Once you see `normal termination` and a reasonable structure, record the following in section 3W-a of your `scratch.tex`:

- the xTB version (`xtb --version`)
- the OpenBabel version (`obabel --version`)
- one sentence confirming the test optimisation completed normally

You are ready to proceed to Part 2.